

English-to-Venetian Machine Translation

A comparison of NLLB-200 zero-shot translation, an LLM zero-shot baseline, and the best fine-tuned NLLB configuration using automatic metrics and human ratings on a 1–5 scale.

1. Dataset

Training examples

653

Development examples

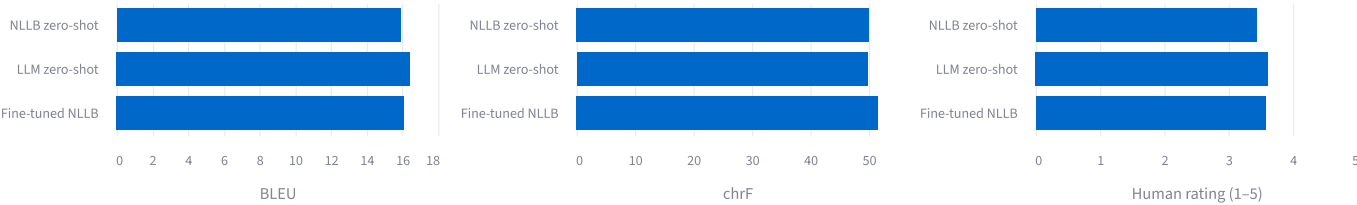
82

Test examples

82

2. Aggregate test results

System	BLEU	chrF	Human rating (1–5)	Sentences
NLLB zero-shot	15.89	49.97	3.43	82
LLM zero-shot	16.43	49.74	3.61	82
Fine-tuned NLLB	16.07	51.50	3.57	82



Difference from NLLB zero-shot

Fine-tuned: BLEU

+0.18

Fine-tuned: chrF

+1.53

Fine-tuned: Human rating

+0.14

LLM: BLEU

+0.54

LLM: chrF

-0.23

LLM: Human rating

+0.18

The fine-tuned NLLB model improves BLEU, chrF, and the mean human rating over the NLLB zero-shot baseline. The LLM zero-shot baseline has the highest BLEU and human rating, while the fine-tuned NLLB model has the highest chrF.

3. Sentence-level comparison

Fine-tuned better than NLLB

44

Fine-tuned worse than NLLB

37

Equal sentence score

1

Example view

☒ Presentation example ☐ Largest fine-tuned improvements ☐ Largest fine-tuned regressions ☐ All test examples

Select a test sentence

0737 — These aren't very nimble so the trail has to take a long swerve around the Transantarctic Mount...

English source

These aren't very nimble so the trail has to take a long swerve around the Transantarctic Mountains to come onto the plateau.

Venetian reference

No i xe tanto agili, quindi l'itinerario el ga da ciapar na longa deviasión intorno ale Montagne Transantartiche pa rivar fin a l'altopiano.

NLLB zero-shot

Questi no xe molto agili, quindi el sentiero ga da far una lunga svolta intorno a le montagne transantartiche per arrivare in altopiano.

Sentence chrF

Human score (1–5)

LLM zero-shot

Sti qua no i xe tanto svelti, cussì el sentier ghe toca far un giro longo intorno ai Monti Transantartici par rivar sul altopiano.

Sentence chrF

Human score (1–5)

Fine-tuned NLLB

Sti no i xe massa agili quindi el sentiero el ga da far na longa svolta intorno a le montagne transantartiche par rivare sul altopiano.

Sentence chrF

Human score (1–5)

46.72

3

37.33

↓ -9.38

3

48.58

↑ +1.87

3

[Technical details](#)

4. Try your own sentence

Enter an English sentence and generate a new Venetian translation at runtime.

[Translation model settings](#)

Translation system

☐ NLLB zero-shot ☒ Fine-tuned NLLB

English text

Type an English sentence here...

Translate into Venetian

Takeaway: the fine-tuned NLLB model improves BLEU, chrF, and the mean human rating over the NLLB zero-shot baseline. The LLM zero-shot baseline obtains the highest BLEU and human rating, while the fine-tuned NLLB model obtains the highest chrF. Mean human ratings are 3.43 for NLLB zero-shot, 3.61 for LLM zero-shot, and 3.57 for fine-tuned NLLB.